DESCRIPTION OF COURSEWORK

Course Code	EEE304
Course Name	High Voltage Engineering
Lecturer	Ir Dr Mohd Faris bin Abdullah
Academic Session	09/2023
Assessment Title	Assignment - Lightning Risk Assessment

A. Introduction/ Situation/ Background Information

Objectives

To conduct the lightning risk assessment using relevant engineering codes and standard.

Pre-Assignment

Knowledge on lightning risk assessment and familiarity with RAPAL software is very important prior to start working on this assignment. Therefore, students should complete the Pre-Assignment steps below:

1. Study the lightning risk assessment lecture notes and IEC standards.
2. Install the relevant software.
3. Learn on how to use the relevant software.

Visit Course Lightning-2: Lightning Protection System Design and Calculations ~ Electrical Knowhow (electrical-knowhow.com) website for additional information on lightning risk assessment and also on using the RAPAL software.

B. Course Learning Outcomes (CLO) covered

At the end of this assessment, students are able to:

CLO 3 Determine the over-voltage phenomena, insulation coordination in power system, types of high voltage testing for electrical apparatus and non-destructive testing of materials.

C. University Policy on Academic Misconduct

1. Academic misconduct is a serious offense in Xiamen University Malaysia. It can be defined as any of the following:
    i. **Plagiarism** is submitting or presenting someone else's work, words, ideas, data or information as your own intentionally or unintentionally. This includes incorporating published and unpublished material, whether in manuscript, printed or electronic form into your work without acknowledging the source (the person and the work).
    ii. **Collusion** is two or more people collaborating on a piece of work (in part or whole) which is intended to be wholly individual and passed it off as own individual work.
    iii. **Cheating** is an act of dishonesty or fraud in order to gain an unfair advantage in an assessment. This includes using or attempting to use, or assisting another to use materials that are prohibited or inappropriate, commissioning work from a third party, falsifying data, or breaching any examination rules.
2. All assessments submitted must be the student's own work, without any materials generated by AI tools, including direct copying and pasting of text or paraphrasing. Any form of academic misconduct, including using prohibited materials or inappropriate assistance, is a serious offense and will result in a zero mark for the entire assessment or part of it. If there is more than one guilty party, such as in case of collusion, all parties involved will receive the same penalty.

D. Instruction to Students

This is a group of **four (4)** students' assignment to be completed in 2 weeks.

One (1) report in softcopy (PDF) should be submitted and follow the following format:

1. Methodology
2. Results and Discussion

Due Date: 7th December 2023 (Week 12)

Late submission will not be accepted.

Software Tools:

1. RAPAL

E. Evaluation Breakdown

No.	Component Title	Percentage (%)
1.	Identify the risk components and perform manual calculation on lightning risk assessment study	50
2.	Perform simulation on lightning risk assessment study	40
3.	Compare and discuss the manual calculation and simulation results	10
	TOTAL	**100**

F. Task(s)

You just join an electrical consultant company. As part of the training, your mentor instructs you to conduct lightning risk assessment of any existing bungalow house around by neglecting the existing lightning protection system (if available).

1. Perform lightning risk assessment study
    a. Choose a standalone bungalow building around you and take pictures of all sides of the building covering adjacent structures. Download from google map the aerial satellite view for your analysis. Use your best judgement for the dimensions of the building.
    b. Evaluation must be based on IEC 62305 standard.
    c. Perform manual calculation on lightning risk assessment study.
    d. Perform simulation on lightning risk assessment study using RAPAL software.
    e. Compare and discuss the manual calculation with simulation results.
    f. Display the simulation snapshot results in your report.
2. Present your lightning risk assessment layout in 2D drawing with proper engineering drawing quality (legend and symbol, title block, etc.)

G. Description of Complex Engineering Problem

	Attributes	Remarks
WP1	**In-depth Engineering Knowledge Required** (Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach)	This assignment requires students to possess in-depth knowledge of lightning risk assessment using relevant engineering codes and standard. Students need to assess the lightning risk by performing calculation using fundamental knowledge (WK3). It may encompass some elements of WK4, WK5 and WK6
WP2	**Wide Range of Conflicting Requirement** (Involve wide-ranging or conflicting technical, engineering and other issues)	In assessing the lightning risk, students have to be very well verse with relevant IEC standards. Students need to use their judgement in determining the conflicting input parameters to be used in the lightning risk assessment study
WP3	**Depth of Analysis Required** (No obvious solution and require abstract thinking, originality in analysis to formulate suitable models)	This assignment is an open-ended in nature that can lead to a few possible lightning risk assessment results. Students have to carry out the lightning risk assessment study on an actual bungalow building
WP4	**Familiarity of Issues** (Involve infrequently encountered issues)	Depending on the unique location of the bungalow, students may encounter complex situation of assessing the required risk components to be adopted in the lightning risk assessment study
WP5	**Extent/Beyond use of Applicable Codes/Standard** (Outside problems encompassed by standards and codes of practice for professional engineering)	Lightning risk assessment study on an actual bungalow may demand students to seek information beyond the IEC Standard requirement
WP6	**Extent/Wide Ranging of Stakeholder and Level of Conflicting Requirement** (Diverse groups of stakeholders with widely varying needs)	
WP7	**Interdependence**	

	(High level problems including many component parts or sub-problems)	

APPENDIX 1

MARKING RUBRICS

Component Title	**Lightning Risk Assessment**				**Percentage (%)**		
Criteria	**Score and Descriptors**					**Weight (%)**	**Marks**
	Excellent (5)	**Good (4)**	**Average (3)**	**Need Improvement (2)**	**Poor (1)**		
Identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building	100% detail identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building according to relevant IEC standards with comprehensive discussion	80% detail identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building according to relevant IEC standards with comprehensive discussion	60% detail identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building according to relevant IEC standards with comprehensive discussion	40% detail identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building according to relevant IEC standards with comprehensive discussion	Less than 40% detail identifying the risk components and perform manual calculation on lightning risk assessment study of a standalone bungalow building according to relevant IEC standards with comprehensive discussion	10	
Perform simulation on lightning risk assessment study	100% detail lightning assessment simulation conducted according to relevant IEC standards with comprehensive discussion	80% detail lightning assessment simulation conducted according to relevant IEC standards with comprehensive discussion	60% detail lightning assessment simulation conducted according to relevant IEC standards with comprehensive discussion	40% detail lightning assessment simulation conducted according to relevant IEC standards with comprehensive discussion	Less than 40% detail lightning assessment simulation conducted according to relevant IEC standards with comprehensive discussion	8	
Compare and discuss the manual calculation and simulation results	100% detail comparison between manual calculation and simulated results with comprehensi	80% detail comparison between manual calculation and simulated results with comprehensi	60% detail comparison between manual calculation and simulated results with comprehensi	40% detail comparison between manual calculation and simulated results with comprehensi	less than 40% detail comparison between manual calculation and simulated results with	2	

	ve discussion	ve discussion	ve discussion	ve discussion	comprehensive discussion		
						TOTAL	

Note to students: Please print out and attach this appendix together with the submission of coursework

APPENDIX 2

COMPLEX ENGINEERING PROBLEMS

- Context and definitions based on Graduate Attributes and Professional Competencies v3, published on 21 June 2013 by International Engineering Alliance.
- Complex Engineering Problems have characteristic of **WP1 and some or all of WP2 to WP7**.

Complex Problem Solving (WP)

No.	Attribute	Complex Problem Solving (WP)
WP1	**Depth of knowledge required**	Cannot be resolved without in-depth engineering knowledge at the level of one or more of WK3, WK4, WK5, WK6 or WK8 which allows a fundamental-based, first principles analytical approach
WP2	**Range of conflicting requirements**	Involve wide-ranging or conflicting technical, engineering and other issues
WP3	**Depth of analysis required**	Have no obvious solution and require abstract thinking, originality in analysis to formulate suitable models
WP4	**Familiarity of issues**	Involve infrequently encountered issues
WP5	**Extent of applicable codes**	Are outside problems encompassed by standards and codes of practice for professional engineering
WP6	**Extent of stakeholder involvement and level of conflicting requirements**	Involve diverse groups of stakeholders with widely varying needs
WP7	**Interdependence**	Are high level problems including many component parts or sub-problems

Knowledge Profile (WK)

No.	Attribute	Knowledge Profile (WK)
WK1	**Natural Sciences**	A systematic, theory-based understanding of the natural sciences applicable to the discipline
WK2	**Mathematics**	Conceptually-based mathematics, numerical analysis, statistics and formal aspects of computer and information science to support analysis and modelling applicable to the discipline
WK3	**Engineering Fundamentals**	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline
WK4	**Specialist Knowledge**	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline
WK5	**Engineering Design**	Knowledge that supports engineering design in a practice area
WK6	**Engineering Practice**	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline
WK7	**Comprehension**	Comprehension of the role of engineering in society and identified issues in engineering practice in the discipline: ethics and the professional responsibility of an engineer to public safety; the impacts of engineering activity: economic, social, cultural, environmental and sustainability
WK8	**Research Literature**	Engagement with selected knowledge in the research literature of the discipline